

Supplement to “Eco-evolutionary deviations from steady state across the Hawaiian chronosequence from a maximum entropy perspective”

This supplement walks through the steps of recreating the entire analysis presented in the main text. It also provides further analysis in support of and to contextualize results in the main text.

Computing setup

Analyses are supported by a custom R package *HawaiiArthMETE* [foo] which can be installed from github.

```
devtools::install_github("ecoevomatics/HawaiiArthMETE")
```

The quarto [quarto] document used to generate this supplement can be found with the R command `system.file("ms/hawaii-arth-mete-supp.qmd", package = "HawaiiArthMETE")`.

The following R computing environment is used for all analyses:

```
library(meteR)
library(ggplot2)
library(ggh4x)
library(dplyr)
```

```
Attaching package: 'dplyr'
```

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)
library(vegan)
```

Loading required package: permute

```
library(metafor)
```

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

Loading required package: metadat

Loading required package: numDeriv

Loading the 'metafor' package (version 5.0-1). For an introduction to the package please type: help(metafor)

Attaching package: 'metafor'

The following object is masked from 'package:vegan':

permutest

```
library(HawaiiArthMETE)
```

Attaching package: 'HawaiiArthMETE'

The following object is masked from 'package:meterR':

arth

```
knitr::opts_chunk$set(  
  # default figure dims  
  fig.width = 5.5, fig.height = 4,  
  # caching is default for all subsequent chunks  
  cache = TRUE  
)  
  
# load data  
data(arth)  
  
sessionInfo() |>  
  print(locale = FALSE)
```

R version 4.5.2 (2025-10-31)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.0.1

Matrix products: default

BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework

LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] HawaiiArthMETE_0.1.0	metafor_5.0-1	numDeriv_2016.8-1.1
[4] metadat_1.6-0	Matrix_1.7-4	vegan_2.7-2
[7] permute_0.9-8	tidyr_1.3.1	dplyr_1.2.1
[10] ggh4x_0.3.1	ggplot2_4.0.1	meteR_1.2

loaded via a namespace (and not attached):

[1] gtable_0.3.6	jsonlite_2.0.0	compiler_4.5.2	tidyselect_1.2.1
[5] parallel_4.5.2	cluster_2.1.8.1	splines_4.5.2	scales_1.4.0
[9] yaml_2.3.10	fastmap_1.2.0	lattice_0.22-7	R6_2.6.1
[13] generics_0.1.4	knitr_1.50	MASS_7.3-65	tibble_3.3.0
[17] pillar_1.11.0	RColorBrewer_1.1-3	rlang_1.2.0	mathjaxr_2.0-0
[21] xfun_0.52	S7_0.2.1	cli_3.6.5	mgcv_1.9-3
[25] withr_3.0.2	magrittr_2.0.4	digest_0.6.39	grid_4.5.2
[29] rstudioapi_0.17.1	nlme_3.1-168	lifecycle_1.0.5	vctrs_0.7.3
[33] evaluate_1.0.4	glue_1.8.0	farver_2.1.2	rmarkdown_2.29
[37] purrr_1.1.0	tools_4.5.2	pkgconfig_2.0.3	htmltools_0.5.8.1

METE calculations

Comparison of the observed data to predictions from the Maximum Entropy Theory of Ecology (METE) is performed with package *meteR* [rominger-merow2016]. The following code records information about each sample (name, flow age, proportion non-native presence) as well as several metrics of deviation between theory and observation:

- a z^2 value measuring a normalized difference in likelihoods between observed data and theoretical expectation
- differences in proportion of rarity (number of singletons or normalized metabolic rate ≤ 2) observed and expected
- differences in proportion of dominance (sum of top three most abundant species or sum of top three highest normalized metabolic rates) observed and expected

All metrics are calculated for both the species abundance distribution (SAD) and individual metabolic rate (or power) distribution (IPD).

```
arth_mete <- split(arth, arth[, c("site", "tree")], drop = TRUE) |>
  lapply(function(x) {
    # the esf for this grouping
    esf <- meteESF(spp = x$species_code,
                   abund = x$abundance,
                   power = x$metabolic_rate)

    # sad and ipd
    s <- sad(esf)
    p <- ipd(esf)

    data.frame(
      # sample info
      site = x$site[1],
```

```

    site_name = x$site_name[1],
    tree = x$tree[1],
    flow_age_my = x$flow_age_my[1],

    # z2 values
    z2_sad = logLikZ(s)$z,
    z2_ipd = logLikZ(p)$z,

    # prop non-native for abundance and power
    prop_n_non_nat = sum(
      x$abundance[!(x$origin %in% c("endemic", "indigenous"))]
    ) /
      sum(x$abundance),
    prop_p_non_nat = sum(
      x$metabolic_rate[!(x$origin %in% c("endemic", "indigenous"))]
    ) /
      sum(x$metabolic_rate),

    # differences in extreme values
    n1_diff = log(mean(meteDist2Rank(s) == 1) / mean(s$data == 1)),
    nmax_diff = log(sum(meteDist2Rank(s)[1:10]) / sum(s$data[1:10])),
    p1_diff = log(mean(meteDist2Rank(p) <= 3) / mean(p$data <= 3)),
    pmax_diff = log(sum(meteDist2Rank(p)[1:10]) / sum(p$data[1:10]))
  )
})

arth_mete <- do.call(rbind, arth_mete)

outliers <- filter(arth_mete, nmax_diff < -300)

names(mete_by_non)
names(arth_mete)
arth_mete <- full_join(arth_mete, mete_by_non, by = c("site", "tree"))

select(arth_mete, site_name.x, prop_nspp_non_nat, nmax_diff.x, z2_sad.x) |>
  # pivot so both metrics are plotted in faceted panels
  pivot_longer(prop_nspp_non_nat:nmax_diff.x, names_to = "metric") |>
  ggplot(aes(value, z2_sad.x, color = site_name.x)) +
  geom_point() +

  # add arrows pointing to outliers
  geom_segment(

```

Table 1

```
filter(arth, tree == outliers$tree[1] & site == outliers$site[1]) |>
  summarize(n = n(), .by = c(genus, species, origin)) |>
  arrange(desc(n)) |>
  head(3) |>
  knitr::kable(
    caption = paste(
      "Top three most abundant taxa from",
      levels(arth$site_name)[as.numeric(outliers$site[1])],
      "outlier sample"
    )
  )
```

```
data = pivot_longer(outliers,
  c(prop_n_non_nat, nmax_diff),
  names_to = "metric"),
aes(x = value * 0.8, y = z2_sad * 0.9,
  xend = value * 0.97, yend = z2_sad * 0.99),
arrow = arrow(length = unit(1, "char")),
inherit.aes = FALSE
) +

# facet by metric and trick strip labels into being
# axis labels
facet_grid(
  cols = vars(metric), scales = "free_x",
  labeller = as_labeller(
    c(nmax_diff.x = "Diff. in maximum abundance",
      prop_nspp_non_nat = "Prop. non-native individuals")
  ),
  switch = "x"
) +
ylab(expression("SAD"~z^2*"-value")) +
this_ms_look() +
theme(strip.background = element_blank(),
  strip.placement = "outside",
  axis.title.x = element_blank())
```

Table 2

```
filter(arth, tree == outliers$tree[2] & site == outliers$site[2]) |>
  summarize(n = n(), .by = c(genus, species, origin)) |>
  arrange(desc(n)) |>
  head(3) |>
  knitr::kable(
    caption = paste(
      "Top three most abundant taxa from",
      levels(arth$site_name)[as.numeric(outliers$site[2])],
      "outlier sample"
    )
  )
```

```
crazy_abund <- filter(arth,
                      (site %in% outliers$site & tree %in% outliers$tree)) |>
  summarize(n = n(), .by = c(site, tree, species_code)) |>
  arrange(desc(n)) |>
  head(3)

summarize(arth, .by = c(site, tree),
          nmax = max(table(species_code))) |>
  filter(nmax < 300) |>
  ggplot(aes(nmax)) +
  geom_histogram(bins = 20) +
  geom_segment(data = crazy_abund,
              mapping = aes(x = n, y = rep(10, 3),
                           xend = n, yend = rep(0, 3)),
              color = "red",
              arrow = arrow(length = unit(1, "char")),
              inherit.aes = FALSE) +
  xlab("Maximum abundances") +
  ylab("Count") +
  this_ms_look()
```

```
# remove from mete results
arth_mete_out <- arth_mete
arth_mete <- filter(arth_mete,
                    !((site %in% outliers$site) & (tree %in% outliers$tree)))
```

```
# remove from arth data frame
arth_out <- arth
arth <- filter(arth,
               !((site %in% outliers$site) & (tree %in% outliers$tree)))
```

Deviations from METE across the chronosequence

Now we analyze how deviations from METE behave across the chronosequence. The following code produces the analysis of a hump-shaped versus flat relationship between METE deviation and flow age.

```
# quadratic model of sad z^2 across age
sad_rma_chrono <- site_means_lm(log(z2_sad) ~
                                log(flow_age_my) + I(log(flow_age_my)^2) +
                                (1 | site),
                                data = arth_mete)

# quadratic model of ipd z^2 across age
ipd_rma_chrono <- site_means_lm(log(z2_ipd) ~
                                log(flow_age_my) + I(log(flow_age_my)^2) +
                                (1 | site),
                                data = arth_mete)
```

Likelihood ratio tests (LRT) can evaluate the significance of the quadratic model

```
# likelihood ratio tests for sad and ipd across chrono
sad_lrt_chrono <- anova(sad_rma_chrono)
sad_lrt_chrono
```

Test of Moderators (coefficients 2:3):
 QM(df = 2) = 27.0082, p-val < .0001

```
ipd_lrt_chrono <- anova(ipd_rma_chrono)
ipd_lrt_chrono
```

Test of Moderators (coefficients 2:3):
 QM(df = 2) = 3.5442, p-val = 0.1700

The quadratic model for SAD z^2 -values is supported by the LRT with a P -value < 0.0001 , but the quadratic model for IPD z^2 -values did not have a significant P -value. However, the 95% CI for the quadratic term in the IPD z^2 -values model has the majority of its range below zero, indicating the data do tend toward a quadratic shape, just with greater variance compared to the quadratic model for SAD z^2 -values.

```
ipd_rma_chrono
```

Fixed-Effects with Moderators Model (k = 5)

I² (residual heterogeneity / unaccounted variability): 0.00%
H² (unaccounted variability / sampling variability): 0.25
R² (amount of heterogeneity accounted for): 75.54%

Test for Residual Heterogeneity:
QE(df = 2) = 0.4939, p-val = 0.7812

Test of Moderators (coefficients 2:3):
QM(df = 2) = 3.5442, p-val = 0.1700

Model Results:

	estimate	se	zval	pval	ci.lb	ci.ub
intrcpt	-0.0416	0.4337	-0.0959	0.9236	-0.8917	0.8085
log(flow_age_my)	-0.4574	0.2435	-1.8785	0.0603	-0.9347	0.0198 .
I(log(flow_age_my)^2)	-0.0591	0.0327	-1.8068	0.0708	-0.1231	0.0050 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The following code produces the visualization reported in the main text of deviation from METE across flow ages.

```
# prepare data frame of predicted values for plotting
x <- range(arth$flow_age_my) |>
  log() |>
  (\(x) seq(x[1], x[2], length.out = 50))() |>
  exp() |>
  data.frame(flow_age_my = _)

sad_hat <- site_means_predict(sad_rma_chrono, newdata = x)
```

```

ipd_hat <- site_means_predict(ipd_rma_chrono, newdata = x)

z2_chrono_pred <- bind_rows(z2_sad = data.frame(x, sad_hat),
                           z2_ipd = data.frame(x, ipd_hat),
                           .id = "metric") |>

  select(!se) |>
  rename(z2 = pred,
         z2_lo = ci.lb,
         z2_hi = ci.ub) |>
  mutate(across(starts_with("z2"), exp))

# plotting
fig_z2_chrono <- pivot_longer(arth_mete, c(z2_sad, z2_ipd),
                              names_to = "metric", values_to = "z2") |>
  ggplot(aes(flow_age_my, z2, color = site_name)) +
  geom_jitter(width = 0.1) +
  # confidence envelope
  geom_ribbon(data = z2_chrono_pred,
            mapping = aes(x = flow_age_my,
                          ymin = z2_lo,
                          ymax = z2_hi),
            color = "transparent", alpha = 0.4) +
  # central tendency
  geom_line(data = z2_chrono_pred,
           mapping = aes(flow_age_my, z2),
           color = "black") +
  # facet by IPD or SAD and trick strip label into axis label
  facet_grid(
    rows = vars(metric),
    labeller = as_labeller(
      c(z2_ipd = "\"IPD\"~z^2*\"-value\"",
        z2_sad = "\"SAD\"~z^2*\"-value\""),
      label_parsed
    ),
    switch = "y"
  ) +
  this_ms_look(log_x = TRUE, log_y = TRUE) +
  theme(axis.title.y = element_blank(),
        strip.background = element_blank(),
        strip.placement = "outside")

```

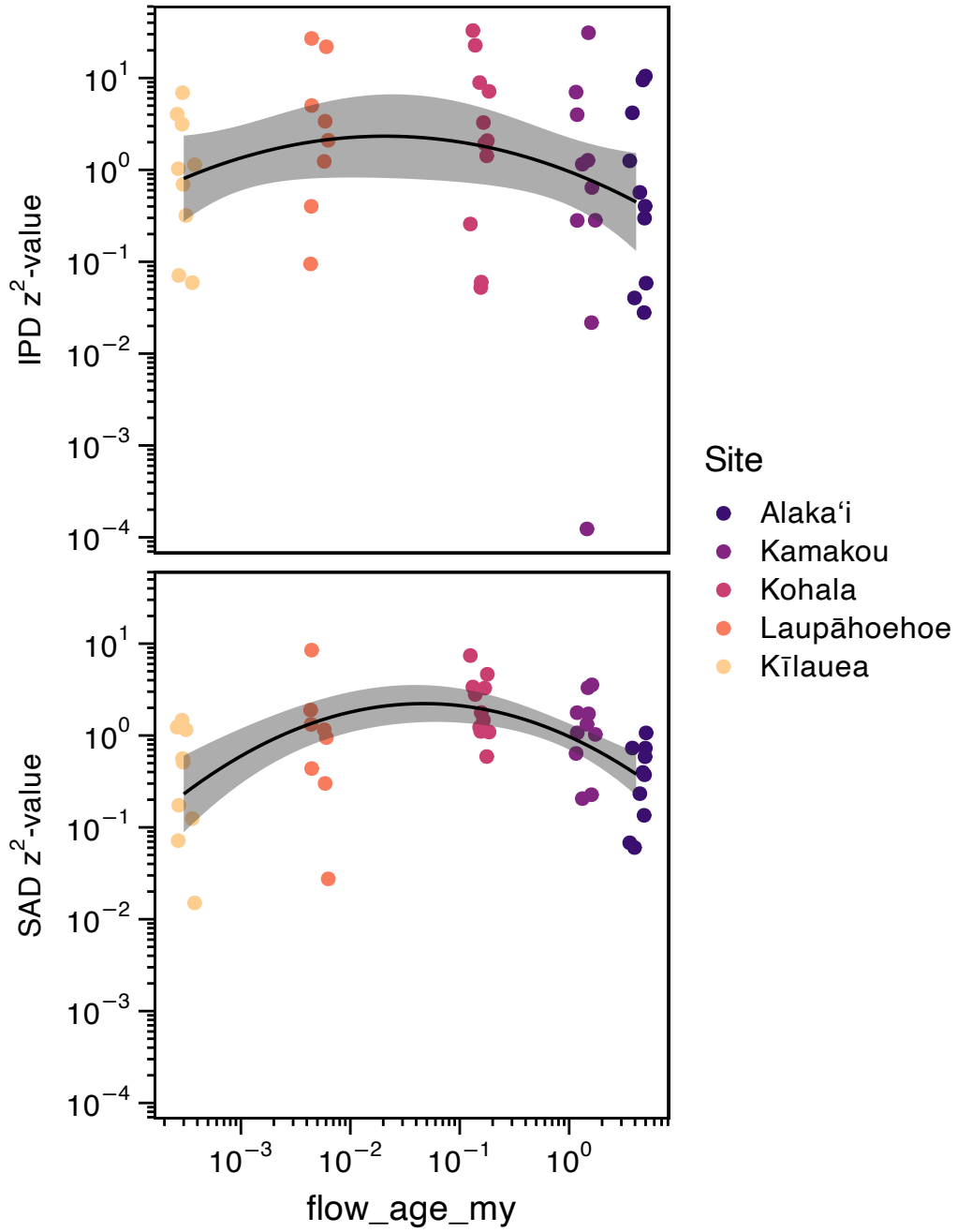


Figure 1: Deviations from METE as measured by z^2 across the chronosequence. Discussed in more detail in the main text.

How do observed patterns deviate from METE?

Next we determine how the observed data deviate from the METE predictions. To begin we look at two exemplar SADs and two exemplar IPDs representing low and high deviation from theory (Figure 2).

```
# function to summarize theoretical and observed distributions
# given one sample `x`

mk_summ <- function(x) {
  esf <- meteESF(x$species_code, x$abundance, x$metabolic_rate)

  list(sad = sad(esf),
       ipd = ipd(esf)) |>
  lapply(function(x) {
    thr <- samp_mete(x, n = 1000) |>
      summarize_samp_mete()

    obs <- data.frame(rank = seq_along(x$data),
                      abund_obs = x$data)

    full_join(thr, obs)
  }) |>
  bind_rows(.id = "dist")
}

expl_samps <- data.frame(site = c("KA", "MO"),
                        tree = c(5, 9),
                        deviation = c("lo", "hi")) |>
  left_join(arth) |>
  (\(x) split(x, x$deviation, drop = TRUE))() |>
  lapply(mk_summ) |>
  bind_rows(.id = "deviation")
```

```
Joining with `by = join_by(site, tree)`
Joining with `by = join_by(rank)`
Joining with `by = join_by(rank)`
Joining with `by = join_by(rank)`
Joining with `by = join_by(rank)`
```

```

fig_how_diff <- ggplot(expl_samps, aes(rank, m)) +
  geom_point(aes(rank, abund_obs)) +
  geom_ribbon(aes(ymin = lo, ymax = hi),
             fill = "#489BC5", alpha = 0.75) +
  geom_line(color = "#489BC5") +
  facet_grid2(
    rows = vars(dist), cols = vars(deviation),
    scales = "free", independent = "x",
    labeller = as_labeller(
      c(hi = "High deviation", lo = "Low deviation",
        ipd = "Metabolic rate", sad = "Abundance")
    ),
    switch = "y"
  ) +
  xlab("Rank") +
  ylab("") +
  this_ms_look(log_y = TRUE) +
  theme(strip.background.y = element_blank(),
        strip.placement.y = "outside",
        strip.text.y = element_text(size = 14))

fig_how_diff

```

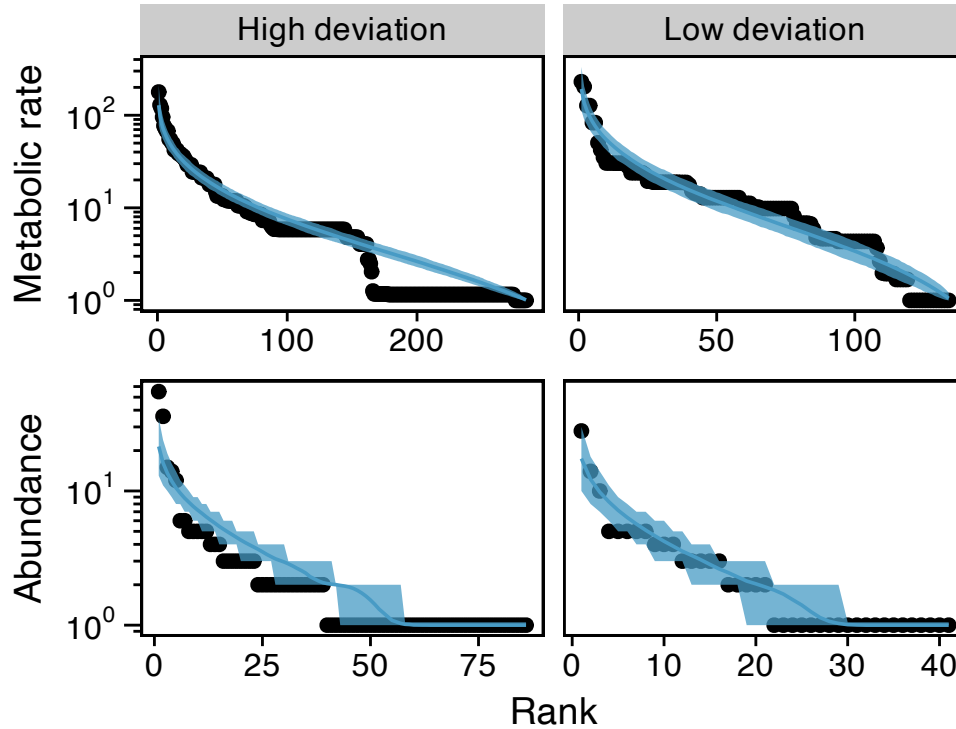


Figure 2: Two exemplar IPDs and SADs showing high and low deviation from METE. The low deviation distributions come from the same sample at the Alaka'i site while the high deviation distributions come from the same sample at the Kamakou site. Samples were chosen to represent ranges of deviation while still being similar in state variables

Looking across all samples (Figure 3) we see that deviations between observed and theoretical SADs is consistently in the direction of under-predicted both rarity and dominance (i.e. observed SADs have more rarity and greater dominance than theoretical predictions). **fig-sad-how** shows that when METE predictions deviate from observed SADs, METE under-predicts both rarity and dominance compared to observed SADs. This means that observed SADs are more uneven than the METE prediction.

```
fig_how_z2 <- bind_rows(sad = select(arth_mete, site_name,
                                   z2 = z2_sad, rar_diff = n1_diff, dom_diff = nmax_diff),
  ipd = select(arth_mete, site_name,
               z2 = z2_ipd, rar_diff = p1_diff, dom_diff = pmax_diff),
  .id = "dist") |>
  pivot_longer(ends_with("diff"),
               names_to = c("comp", ".value"),
               names_sep = "_") |>
```

```

ggplot(aes(diff, z2, color = site_name)) +
  geom_point() +
  facet_grid2(rows = vars(dist), cols = vars(comp),
    scales = "free", independent = "x",
    labeller = as_labeller(
      c(ipd = "\"IPD\"~z^2*\"-value\"",
        sad = "\"SAD\"~z^2*\"-value\"",
        dom = "\"Diff. in dominance\"",
        rar = "\"Diff. in rarity\""),
      label_parsed
    ),
    switch = "both") +
  xlab("") +
  ylab("") +
  this_ms_look(log_y = TRUE) +
  geom_vline(xintercept = 0, color = "gray") +
  theme(strip.background = element_blank(),
    strip.placement = "outside",
    strip.text.y = element_text(size = 14))

```

fig_how_z2

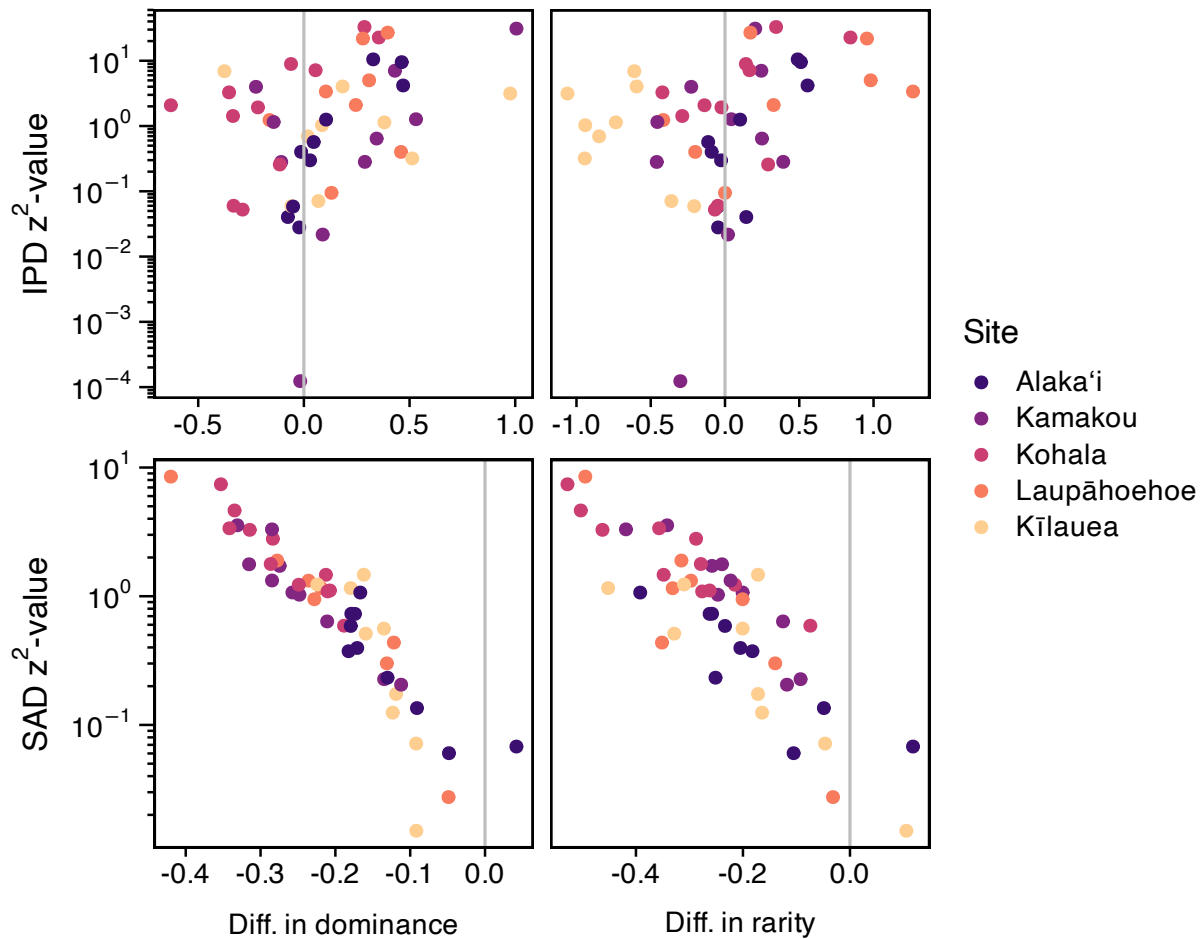


Figure 3: Relationship between z^2 measure of deviation from METE and different metrics describing how observed data and METE predictions diverge.

Comparing deviations from METE with potential explanatory variables

Deviation vs. proportion that are invasive

```
mete_by_non <- summarize(
  arth,
  prop_nspp_non_nat = n_distinct(
    species_code[!(origin %in% c("endemic", "indigenous"))]) /
    n_distinct(species_code),
  .by = site_name) |>
full_join(arth_mete)
```


Joining with `by = join\_by(site\_name)`

```
ggplot(mete_by_non, aes(prop_nspp_non_nat, z2_sad, color = site_name)) +  
  geom_point() +  
  this_ms_look(log_y = TRUE) +  
  geom_hline(yintercept = exp(0.0474))  
  
sad_rma_non <- site_means_lm(log(z2_sad) ~ prop_nspp_non_nat + (1 | site),  
                             filter(mete_by_non, site != "V0"))  
  
anova(sad_rma_non)
```

Test of Moderators (coefficient 2):

QM(df = 1) = 20.0239, p-val < .0001

```
filter(mete_by_non, site == "V0") |>  
  summarize(across(contains("nspp"), mean)) |>  
  site_means_predict(sad_rma_non, newdata = _)
```

pred	se	ci.lb	ci.ub
0.3822	0.1708	0.0474	0.7170

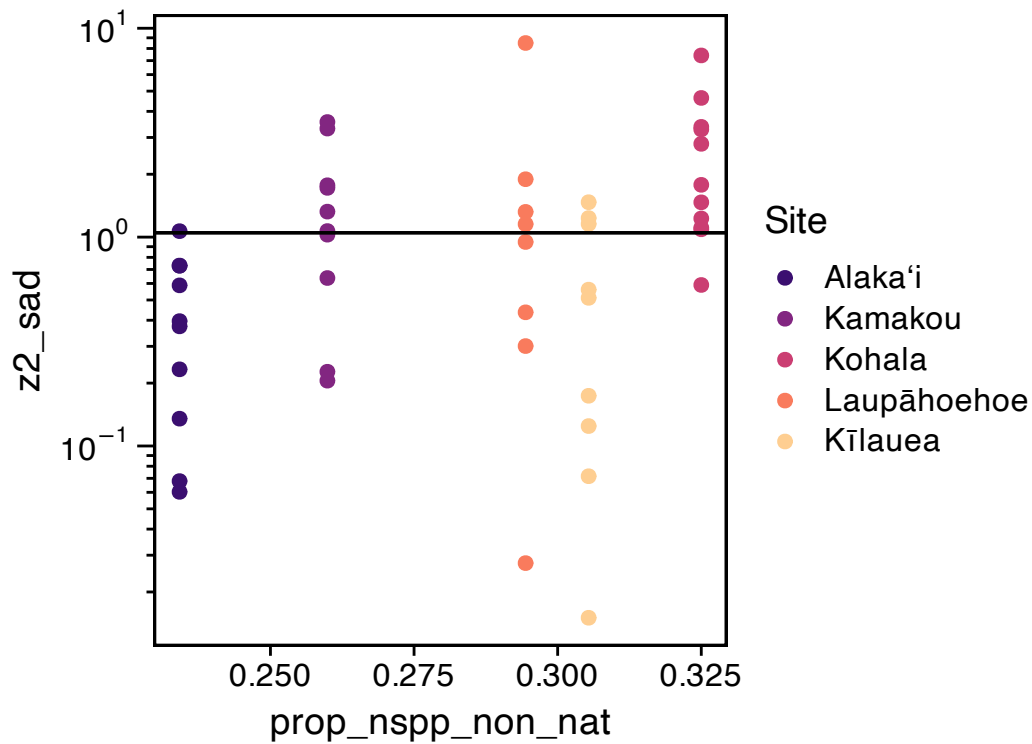


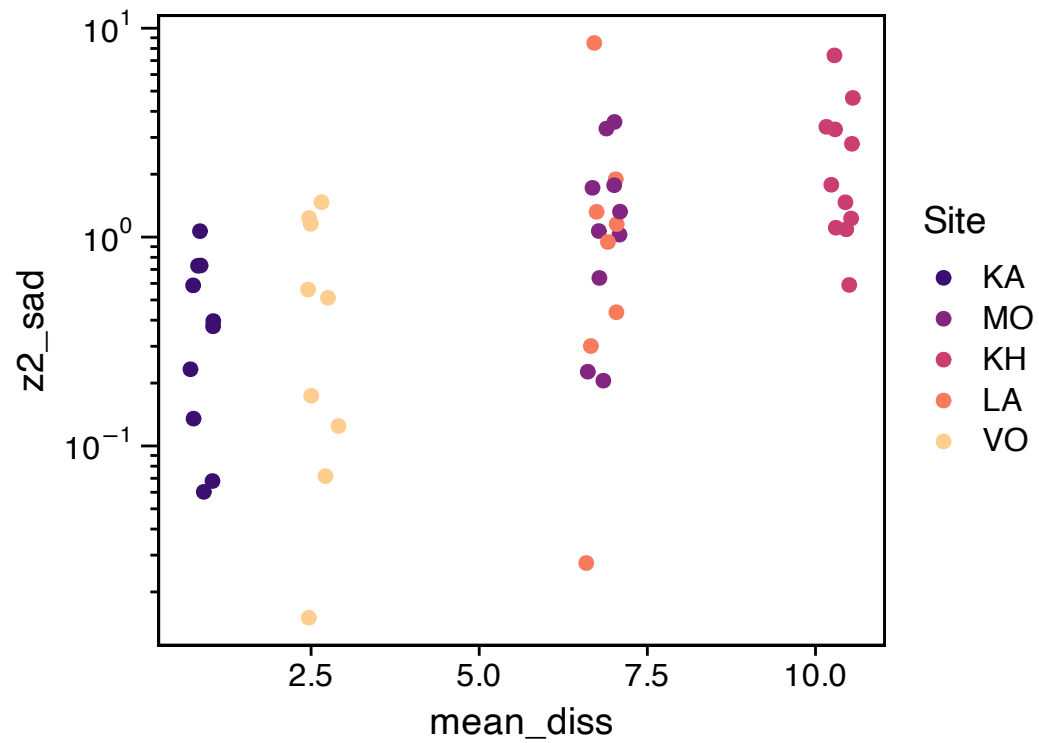
Figure 4

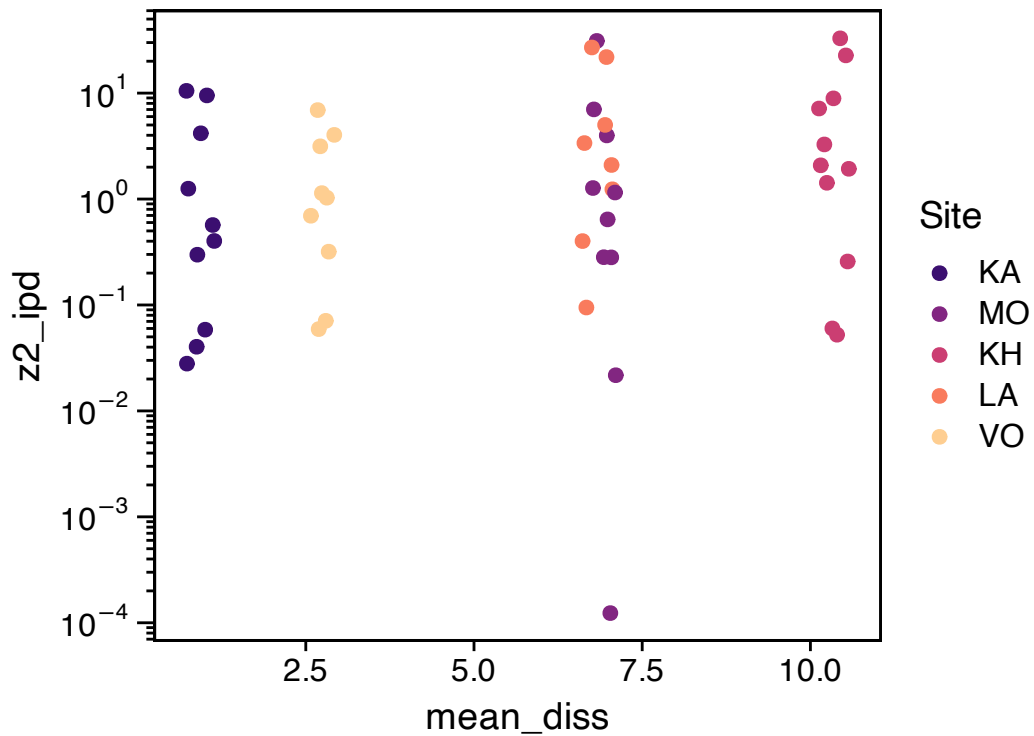
Deviation vs. dissimilarity

```
diss <- split(arth, arth$site) |>
  sapply(FUN = function(dat) {
    summarize(dat, abund = n(), .by = c(tree, species_code)) |>
    pivot_wider(names_from = species_code,
                 values_from = abund,
                 values_fill = 0) |>
    select(!tree) |>
    bray_diss_z()
  })

mete_by_diss <- mutate(arth_mete,
                       mean_diss = diss[site])
```

```
ggplot(mete_by_diss, aes(mean_diss, z2_sad, color = site)) +
  geom_jitter(width = 0.25) +
  this_ms_look(log_y = TRUE)
```





```
xy <- mete_by_diss |>
  summarize(z2_sad_mean = mean(log(z2_sad)),
            z2_sad_sd = sd(log(z2_sad)),
            n = n(),
            mean_diss = first(mean_diss),
            .by = site)

rma(yi = z2_sad_mean,
    vi = z2_sad_sd^2 / n,
    mods = ~ mean_diss,
    data = xy,
    method = "REML")
```

Mixed-Effects Model (k = 5; τ^2 estimator: REML)

τ^2 (estimated amount of residual heterogeneity): 0 (SE = 0.1059)
 τ (square root of estimated τ^2 value): 0
 I^2 (residual heterogeneity / unaccounted variability): 0.00%

H² (unaccounted variability / sampling variability): 1.00
R² (amount of heterogeneity accounted for): 100.00%

Test for Residual Heterogeneity:
QE(df = 3) = 0.8199, p-val = 0.8447

Test of Moderators (coefficient 2):
QM(df = 1) = 28.8968, p-val < .0001

Model Results:

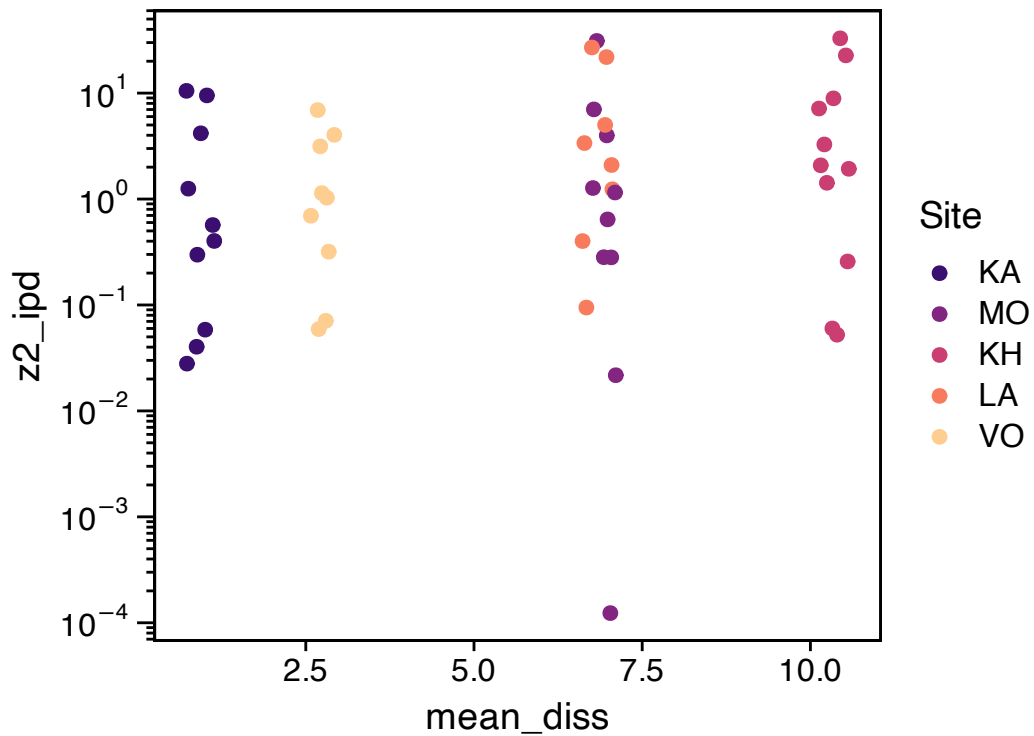
	estimate	se	zval	pval	ci.lb	ci.ub	
intrcpt	-1.5263	0.3095	-4.9311	<.0001	-2.1330	-0.9197	***
mean_diss	0.2173	0.0404	5.3756	<.0001	0.1381	0.2966	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
diss <- split(arth, arth$site) |>
  sapply(FUN = function(dat) {
    summarize(dat, abund = n(), .by = c(tree, species_code)) |>
      pivot_wider(names_from = species_code,
                  values_from = abund,
                  values_fill = 0) |>
      select(!tree) |>
      bray_diss_z()
  })

mete_by_diss <- mutate(arth_mete,
                      mean_diss = diss[site])

ggplot(mete_by_diss, aes(mean_diss, z2_sad, color = site)) +
  geom_jitter(width = 0.25) +
  this_ms_look(log_y = TRUE)
```

```
xy <- mete_by_diss |>
  summarize(z2_sad_mean = mean(log(z2_sad)),
            z2_sad_sd = sd(log(z2_sad)),
            n = n(),
            mean_diss = first(mean_diss),
            .by = site)

rma(yi = z2_sad_mean,
    vi = z2_sad_sd^2 / n,
    mods = ~ mean_diss,
    data = xy,
    method = "REML")
```

Mixed-Effects Model (k = 5; tau² estimator: REML)

```
tau^2 (estimated amount of residual heterogeneity):    0 (SE = 0.1065)
tau (square root of estimated tau^2 value):          0
I^2 (residual heterogeneity / unaccounted variability): 0.00%
```

H² (unaccounted variability / sampling variability): 1.00
R² (amount of heterogeneity accounted for): 100.00%

Test for Residual Heterogeneity:
QE(df = 3) = 0.7632, p-val = 0.8583

Test of Moderators (coefficient 2):
QM(df = 1) = 28.9536, p-val < .0001

Model Results:

	estimate	se	zval	pval	ci.lb	ci.ub	
intrcpt	-1.4856	0.3027	-4.9086	<.0001	-2.0788	-0.8924	***
mean_diss	0.2112	0.0393	5.3809	<.0001	0.1343	0.2882	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
site_bray_z <- split(arth, arth$site) |>
  sapply(function(dat) {
    comm <- summarize(dat,
      abund = n(),
      .by = c(tree, species_code)) |>
    pivot_wider(names_from = species_code,
      values_from = abund,
      values_fill = 0) |>
    select(!tree)
    g <- dat$origin[match(names(comm), dat$species_code)] %in%
      c("endemic", "indigenous")

    bray_part_z(as.matrix(comm), g)
  })
```